# Analysis pipeline: what runs, and why

Every branch in one swing's analysis, as a UML activity diagram. Generated by tools/analysis/pipeline_activity.py and checked against the code.

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## Reading a page

computes · loads from the document · ran: true, did nothing · a step inside a stage

red diamond: a decision the document does not keep. A black one is kept, or partly kept.

bar: fork or join. A join means all of it is finished before what is drawn next; the executor may overlap more, never less.

10 of 19 decisions leave no record in the document. The fine print inside a box lists what it reads (in) and writes (out), from the executor's export.

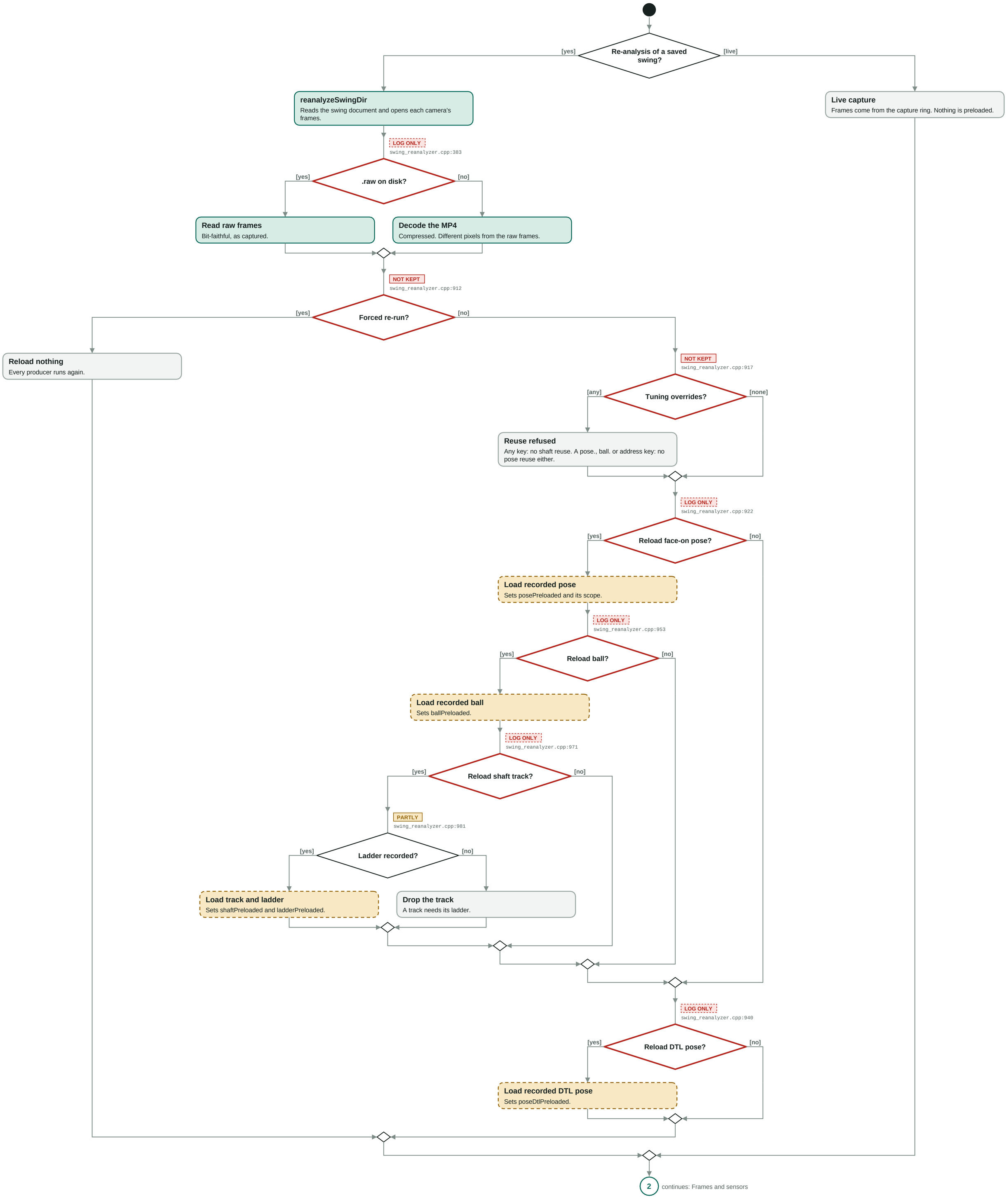
## The decisions

| Decision | Kept? | Where the answer goes | Why it matters |
|---|---|---|---|
| **Forced re-run?** `swing_reanalyzer.cpp:912` | NOT KEPT | Nowhere. The reuse log line sits inside this branch, so a forced run doesn't even log it. | swinglab only; the app never sets it. Throws away every recorded track. |
| **Tuning overrides?** `swing_reanalyzer.cpp:917` | NOT KEPT | Nowhere in the document. swinglab prints one line to stderr. | A pose., ball. or address key re-runs pose. Any key at all re-runs the shaft. |
| **Tracker span-bounded?** `shaft_tracker.cpp:84` | NOT KEPT | Nowhere, and not logged. | --full-window turns it off. Series lengths change; positions don't. |
| **.raw on disk?** `swing_reanalyzer.cpp:383` | LOG ONLY | Log only: an Mp4Reader line appears when the MP4 was read. | Raw frames or the compressed MP4. Everything downstream sees different pixels. |
| **Reload face-on pose?** `swing_reanalyzer.cpp:922` | LOG ONLY | Log only, one [Reanalysis] line. The document says ran: true either way. | Pose version, model file and scope all match, and no pose override. |
| **Reload ball?** `swing_reanalyzer.cpp:953` | LOG ONLY | Log only. ran: true either way. | Needs a reloaded pose and a matching ball version. |
| **Reload shaft track?** `swing_reanalyzer.cpp:971` | LOG ONLY | Log only. ran: true either way. | Reloaded ball, matching shaft version, a recorded track, and NO tuning override of any kind. |
| **Reload DTL pose?** `swing_reanalyzer.cpp:940` | LOG ONLY | Log only. ran: true either way. | Independent of the face-on decision, so a swing can hold a new face-on pose beside an old DTL one. |
| **Coarse pass found a span?** `pose_runner.cpp:776` | LOG ONLY | Log only, one warning. | If not, pass 2 fills the whole window. |
| **DTL witness source?** `analysis_pipeline.cpp:867` | LOG ONLY | Log only: "witness traced" or "from reused flags". | Shifted 2 of 410 published DTL frames on one swing. |
| **Ladder recorded?** `swing_reanalyzer.cpp:981` | PARTLY | Partly: EventRefine and TimelineFusion show ran: false, with no reason. | A track without its ladder cancels shaft reuse. With one, the ladder stages stand down. |
| **Which pose path?** `analysis_pipeline.cpp:605` | PARTLY | Partly: scope says "span" for both bounded and two-pass. Only the log separates them. | The branch #154 was hiding behind. |
| **Implausible samples?** `analysis_pipeline.cpp:848` | PARTLY | Partly: demoted samples carry a flag, not that a reuse pass set it. | The checks run again on every reuse, and can demote one more sample than the fresh run did. |
| **Which ladder wins?** `seg_resolve.h:55` | PARTLY | Partly: phases carry a role and the ladder a confidence. Nothing states which won. | Reused, then adopted sensors, then vision, then the sensor clamp. |
| **Ball found at address?** `shaft_tracker.cpp:272` | PARTLY | Partly: club.addressBall is absent when no ball anchored. | With no ball, a fresh run keeps a synth built before P1's angle was rewritten; a reuse rebuilds it. |
| **Crop accepted?** `pose_runner.cpp:645` | IN DOCUMENT | In the document: pose2d.cropRect, absent when full-frame. | A crop of 90 % of the frame or more, or too few bbox frames, falls back to full-frame. |
| **refine.fusion on?** `analysis_pipeline.cpp:1177` | IN DOCUMENT | In the document: segmentation.fusion. | Picks one of the two ladder builders. On by default. |
| **DTL camera present?** `analysis_pipeline.cpp:2168` | IN DOCUMENT | In the document: the streams list, and stages show ran: false. | Without it the DTL branch stands down, and Skeleton3D starts and refuses. |
| **Motion sensors present?** `analysis_pipeline.cpp:495` | IN DOCUMENT | In the document: the streams list. | Decides the pose path and which ladder wins. |

# 1 How the run starts

Contents

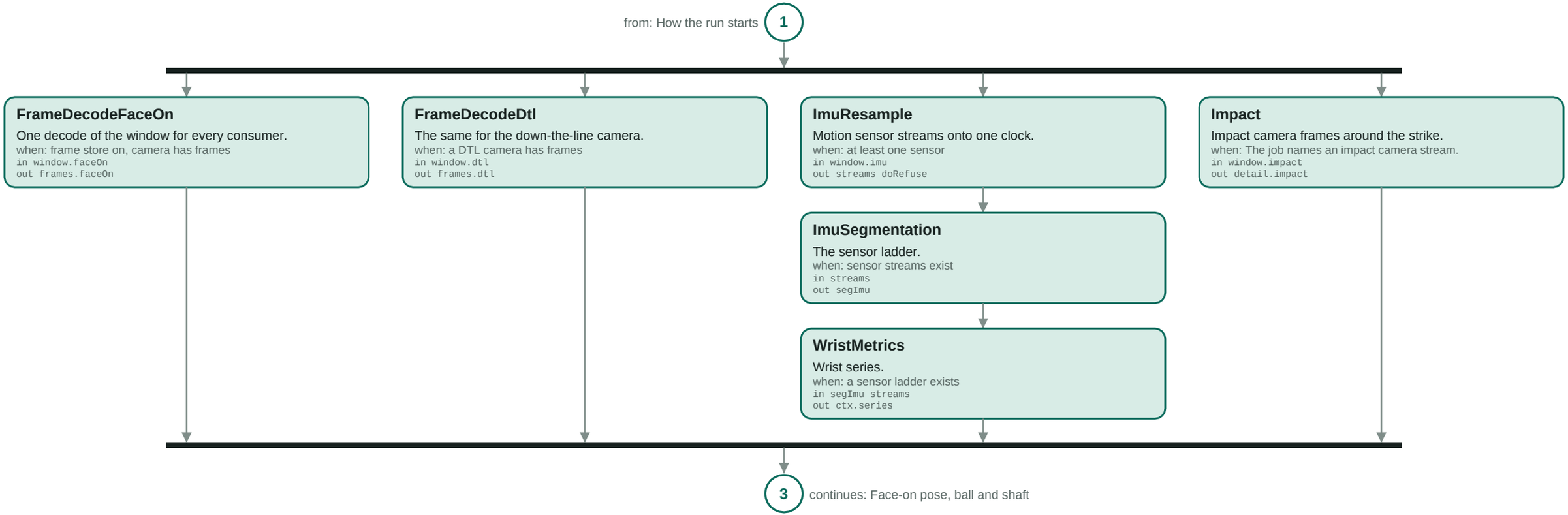
Before any stage runs: what can be reloaded from the swing document. Everything later reads these answers.



# 2 Frames and sensors

Decoded once per camera. The sensors decide the pose path on the next page.

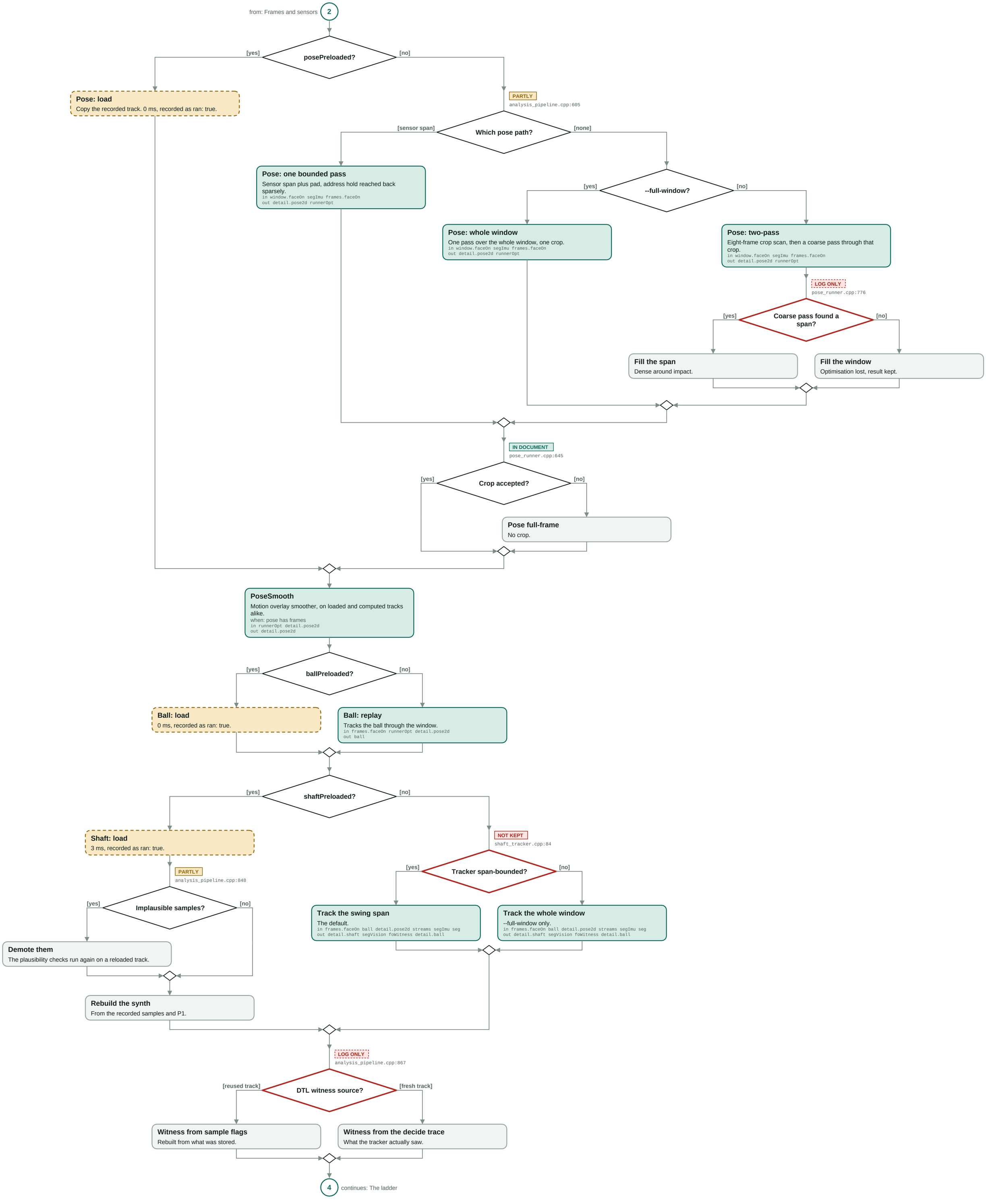
Contents



# 3 Face-on pose, ball and shaft

The heart of #154: each producer either loads or computes, and the document says ran: true both ways.

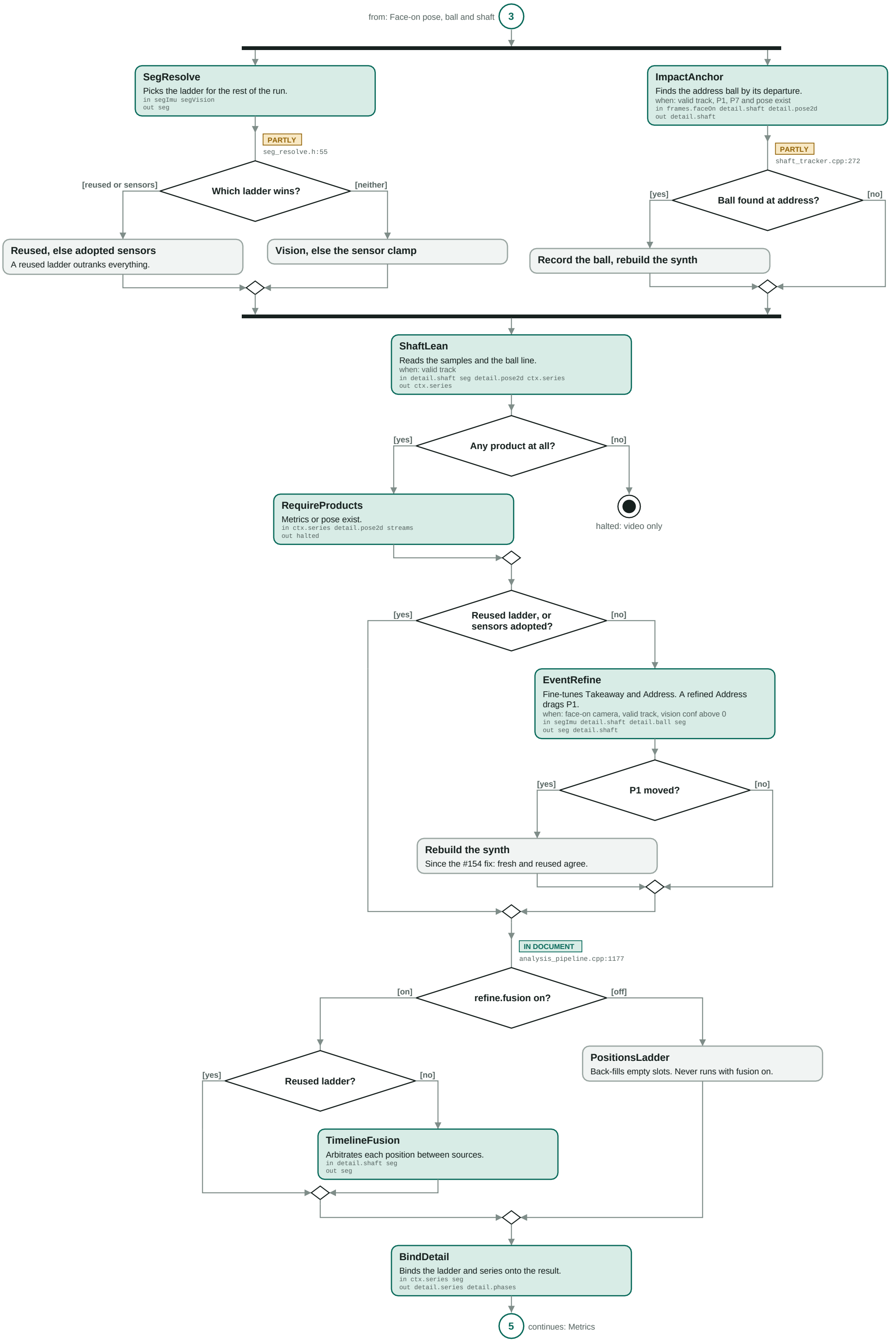
Contents



# 4 The ladder

One ladder of phases and positions. A reused ladder stands the refiners down.

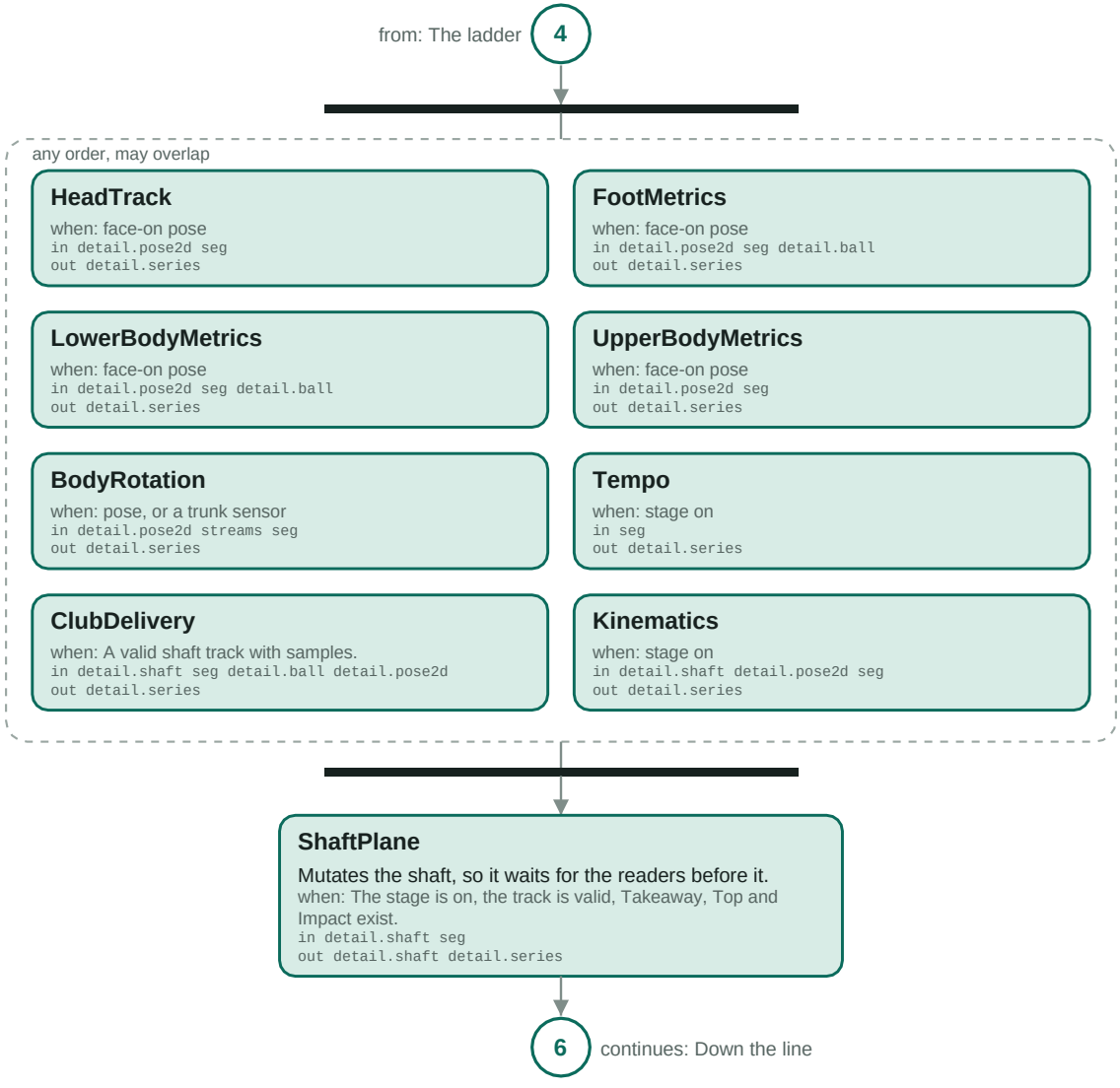
Contents



# 5 Metrics

Contents

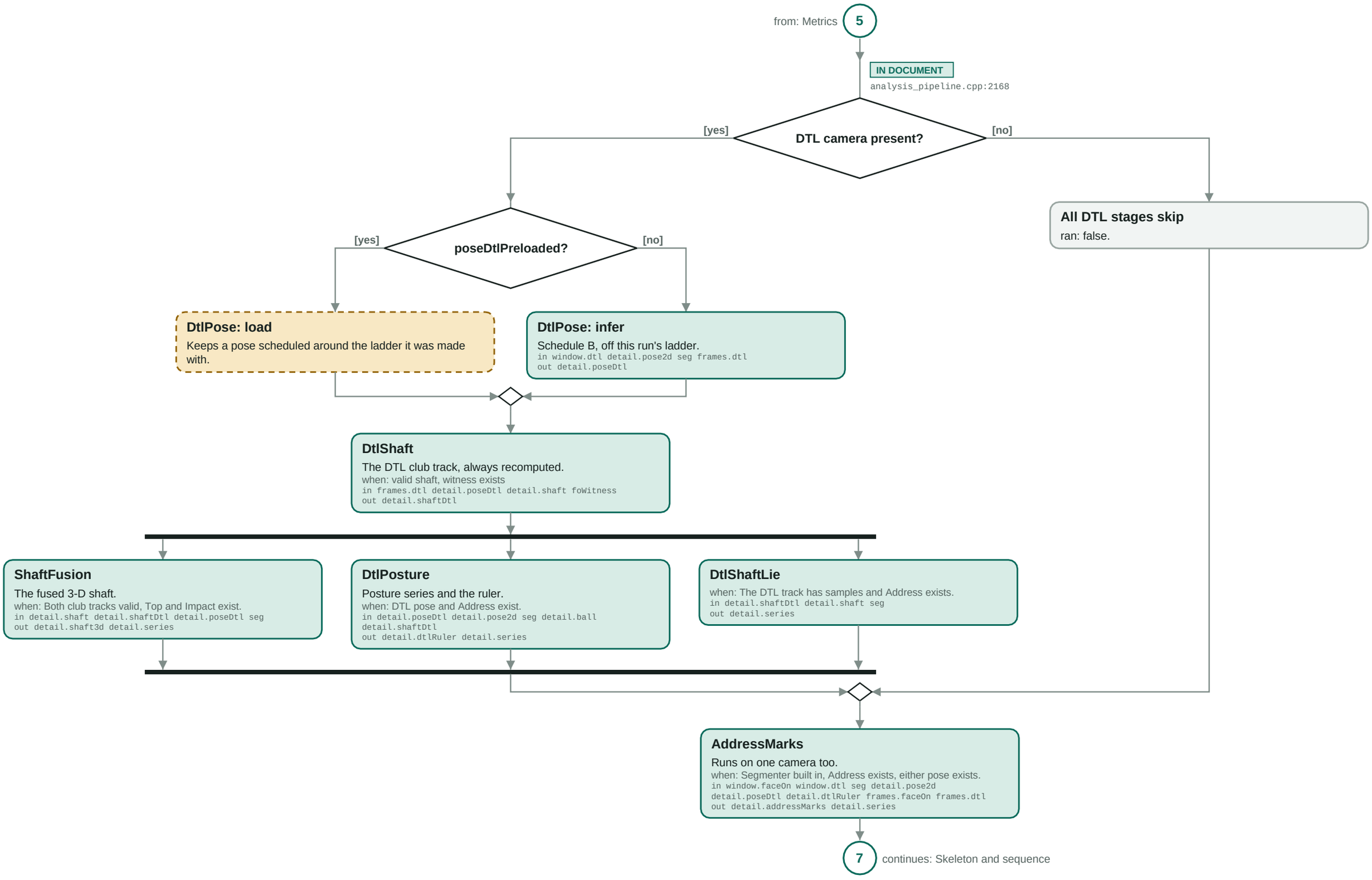
These all append to the series, spliced back in authored order, so they may overlap.



# 6 Down the line

Everything that needs the second camera. Its pose decision is separate from the face-on one.

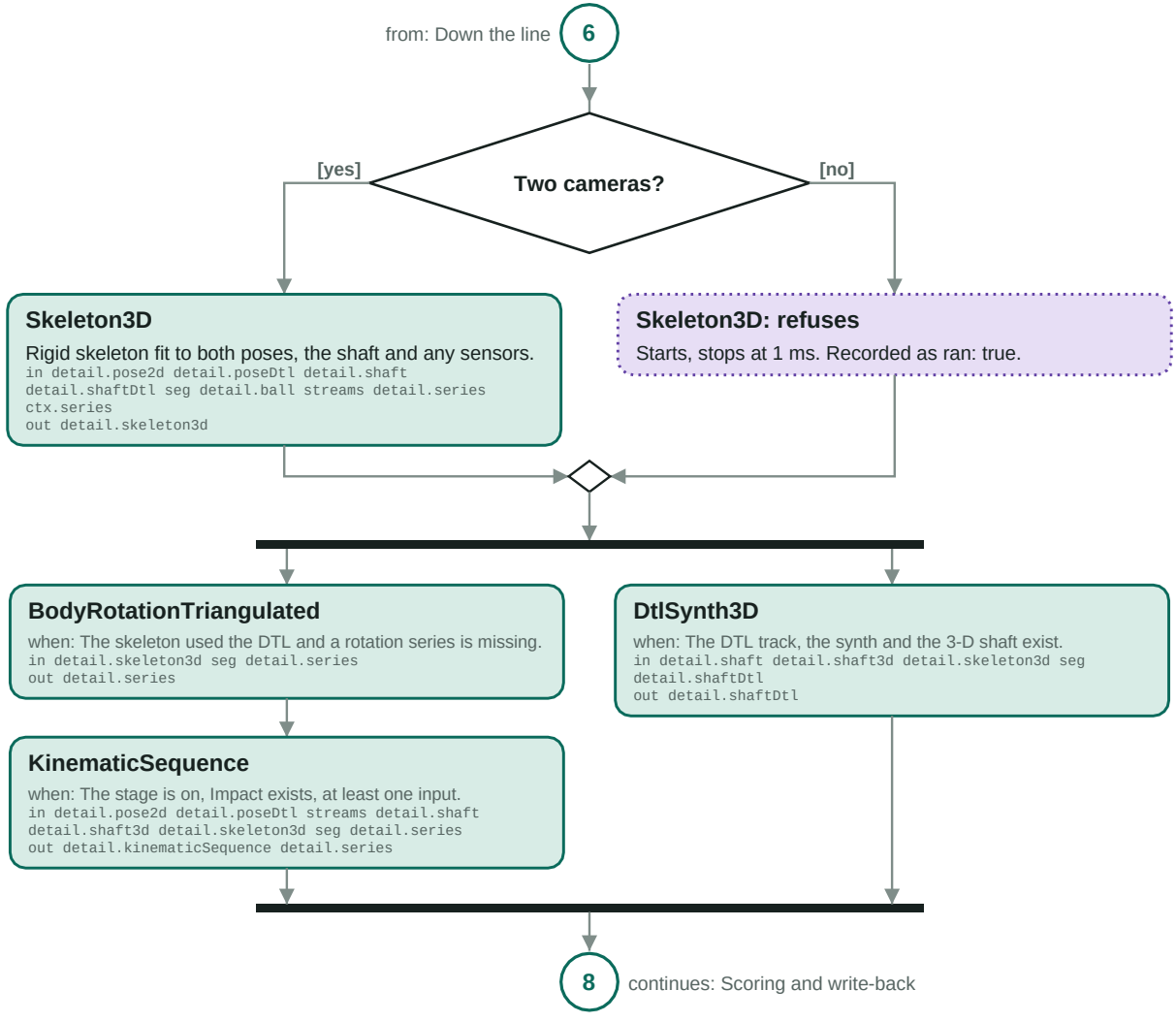
Contents



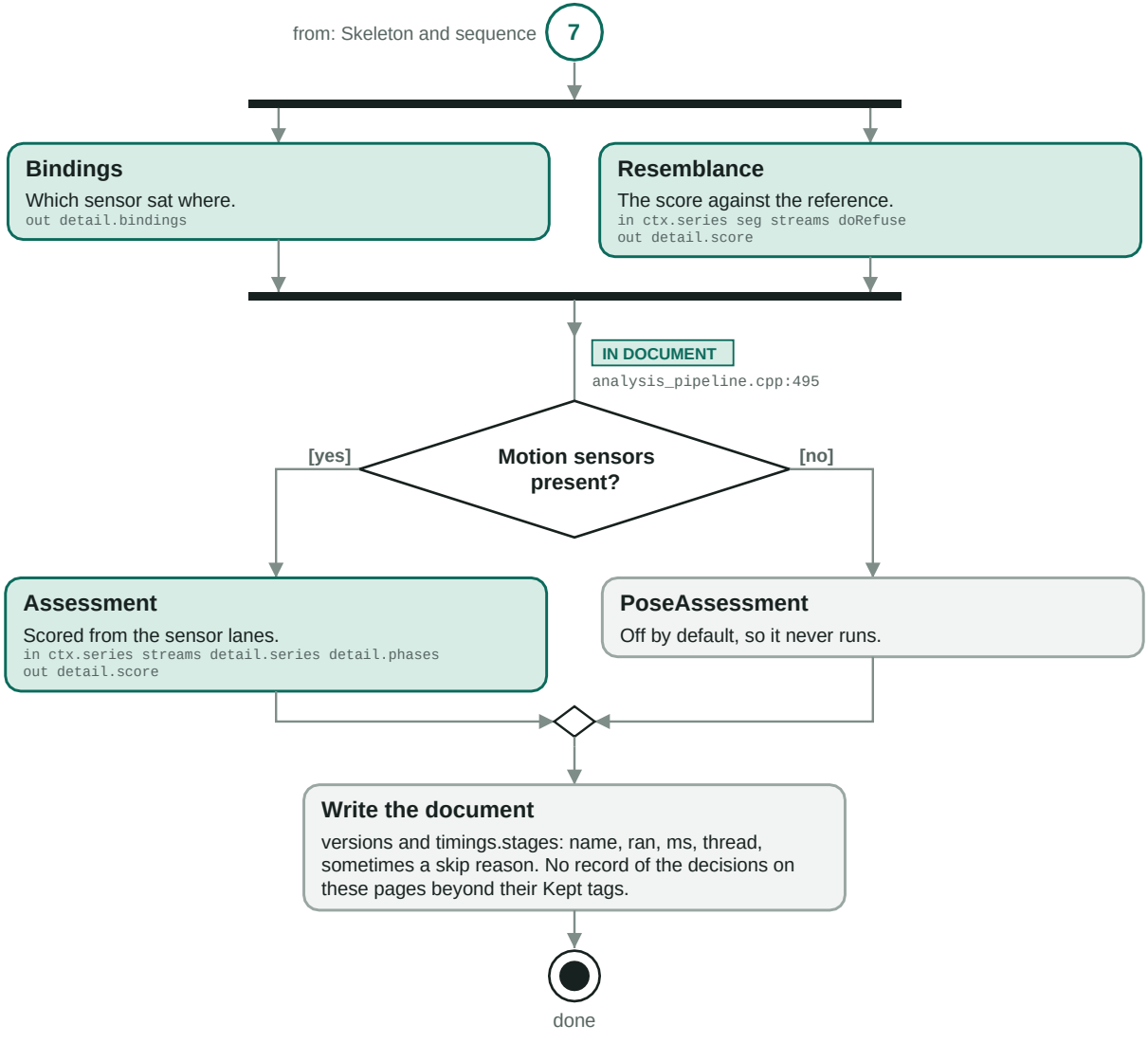
# 7 Skeleton and sequence

Contents

The 3-D fit needs both cameras, but it starts either way.



# 8 Scoring and write-back

The last stages, then what reaches the document.

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